Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/24/2012
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	06/01/2008	**End Date:**	09/05/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
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Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/5/2013 per NSSC information (Ed., 3/10/14) NOTE: Extended to 5/31/14 per NSSC information (Ed., 8/5/13)		
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COI Name (Institution):	Hollingsworth, Andrew (New York University)		
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Task Description:	Colloid science is entering a new era. Over the past 15 years, our NASA-sponsored research has mainly dealt with monodisperse suspensions of colloidal particles interacting via well-known forces. Using spherical particles and observations with light scattering and microscopy, we have gained a great deal of fundamental knowledge about different phases of matter and the dynamics and thermodynamics of their formation. In particular, our experimental results in microgravity have lead to a basic understanding of why crystals and glasses form and their properties. During the past decade, we have made great strides in synthesizing new classes of particles with different shapes and specific, reversible or irreversible, variable range interactions. We have also found new ways to manipulate the particles with flow, electric and magnetic fields and light. We are therefore positioned at the threshold of a new technology, assembling equilibrium and non-equilibrium macroscopic structures with function and activity from well designed particles on the nano to micron scale. Of course, there are still fundamental scientific questions which we can and will address including a host of new ordered phases, frozen configurations, frustration and glasses and the process of self-organization itself. In particular, we plan to use the microscopy and light scattering instruments, in collaboration with our European colleagues, to study particles that we prepare through emulsion and dispersion polymerization. Physical lithographic techniques will also be employed, and the particles will be modified chemically for controllable interactions. We plan to use different phoretic techniques– electro-, dielectro-, and thermo-phoresis– to control the particles density and orientation. These will also serve as the driving forces to establish the rheological properties of these new systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterization of crystal formation in the microgravity environment of the ISS (International Space Station) can lead to a greater understanding of how gravity affects many kinds of colloidal materials, including monodisperse ellipsoids and cubes, colloidal clusters of silica or polymer microspheres, DNA-functionalized colloidal spheres, and 'lock-and-key' colloids. By performing these experiments in reduced gravity, we intend to accomplish the desired characterization without gravitationally-induced inhomogeneities that affect both the dynamics and equilibrium state on earth. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices to be used in optical communication systems. With the ability to make particles of different shapes, i.e., non spherical, we also have the possibility of having directionally dependent particle interactions. For example, we could take tetrahedral clusters of particles and attach DNA to them. The complementary single-stranded DNA 'sticky ends' can associate/dissociate via thermal activation. This arrangement could lead to tetrahedral bonding as found in diamond or in amorphous glass structures. Another approach utilizes depletion interactions. Since we can lithographically prepare particles of any shape we design in two dimensions and many shapes in three dimensions, we can fabricate lock-and-key colloids which only bind to their complementary shape. In this case, the binding is also directional since the congruent surfaces must match. We can also make such lock-and-key particles through emulsion chemistry. Our goal is to produce some simple processes with such ‘designer particles’ and interactions, to lay the foundations for self-assembly and perhaps self-replication of this new class of materials.
	Highlights: • Light Microscopy Module-Dynamic Stage (LMM-DS) Colloids Specialty Cell We provided technical input to Techshot, Inc. — an implementation partner to NASA for enabling space flight experimentation on ISS — to enable them to design a specialty microscope sample cell featuring fine temperature control. This device is critical for systems with temperature controlled interactions such as the DNA-functionalized colloids. Thermally induced hybridization of the constituent linker strands will allow the particles to bind together, thus enabling the colloids to explore self-assembly, reproduction, fidelity, and mutation. • Colloids development (with Dr. S. Sacanna) One of our goals is increased particle fluorescence in confocal microscopy studies of the structure and dynamics of colloidal systems. Sterically stabilized, poly(methyl methacrylate) (PMMA) colloids, core-labeled with rhodamine dye have been the ‘workhorse’ system; however, a time consuming, multi-step synthesis is required to produce them. Furthermore, the dye can diffuse into the non-fluorescent shell such that the core labeling is not sharp and well defined. A promising synthesis from the Sokolov group at Clarkson University was attempted. The resulting ‘ultrabright’ mesoporous silica particles were ~100 nm in size and appeared qualitatively brighter than PMMA colloids using rhodamine 6G. However, the fluorescent particles were colloidally unstable and polydisperse in size. Another approach was based on a method to fabricate colloidal spheres from monodisperse emulsions. The trialkoxysilane monomer (TPM) can undergo hydrolysis and condensation reactions to produce a suspension of monodisperse, micron-sized particles in water. These are charge stabilized by virtue of the dissociation of native silanol groups. Prior to the subsequent free radical polymerization, a 0.002 M RITC dye-coupling agent solution was incorporated into the particles produced by this method. The resulting spherical particles are quantitatively brighter than the PMMA colloids using an epi-fluorescence microscope with a TRITC filter cube. Next, a procedure was developed to transfer the particles into apolar media using the PHS-g-PMMA stabilizer produced in our lab. The final step will be to grow non-fluorescent PMMA or TPM shells around the bright cores and test the core-dyed particles with confocal microscopy. In another particle synthesis effort, micron-sized colloidal cubes were prepared from template iron-oxide (hematite) cubes. Silica shells were grown around the hematite cubes and the iron-oxide cores were subsequently dissolved with a strong acid. The cubes were labeled with the same dye mentioned above. The crystallization behavior of the hollow silica cubes was studied by means of a tunable depletion interaction using temperature sensitive microgel particles as the depletant (see Articles in Peer-reviewed Journals section). Regarding the colloidal cubes, our experiments show that crystallization can be induced with depletion agents (dissolved polymers); however, the micron-size silica cubes are not density matched with the suspending medium. We will study the crystallization and phase behavior of concentrated samples in microgravity. • BCAT-5 results Images of Sample 9, BCAT-5, were downloaded from ISS September 2 and 21, 2011. In this increment, the sample was re-homogenized and the sample was allowed to crystallize. The illumination angle and camera direction were adjusted to

Task Progress:	better image the colloidal crystals. Image analysis is currently underway. • BCAT-6 results In this experiment, the colloidal suspension is contained in a cuvette, half of which is covered by large spherical seeds, while the other portion is not. Two volume fractions were prepared, one at the minimum theoretical volume fraction causing complete crystallization (sample 7) and one in the glassy regime (sample 6). The 'crystallization catalysts' allow us to do several experiments at once. We are seeing the effects of seeding from the cell walls, and determining how the particle volume fraction can affect the rate of crystal nucleation under these conditions. This type of experiment has been complicated because of particle sedimentation on earth. Microgravity allows us to have both index matching (for clarity) and density matching (to avoid sedimentation in 1 g). White light from a flashlight and the camera flash is used to detect the colorful Bragg reflections produced by colloidal crystals formed following the sample mixing. Transcript of March 16, 2012 communication with astronaut Don Pettit indicating that NYU samples 6 and 7 have crystallized. Pettit: "… Station on two for BCAT." KSC: "Go ahead Don. We're here with you on two." Pettit: "I'm in the process, just taking manual pictures of Samples 6 and 7. These guys really look interesting. On Sample 6, there's [sic] no bright scattering centers; however, there is structure in the colloidal haze, particularly notable when I shine the green laser through it. The scattering in the laser beam definitely shows, sort of, macroscopic changes in intensity. And then on Sample number 7, when I shine the green laser through, you can see just these delightful iridescent flakes shining within the laser beam. Really, really neat stuff. I'm going to look forward to working with these two. [omitted text] And I'm taking all these hand held pictures out. Go ahead and put them, when I'm done, in an images for downlink folder called BCAT." KSC: "And Don, BCAT copies all. And, we're really appreciating all the fun you're having. So thank you. Looking forward to those pictures." Pettit: "And I get to have all the fun up here, and you guys get to do all the work." KSC: "We enjoy the work as well." • June 14, 2011 Advanced Colloids Experiment (ACE) Science Concept Review (SCR) The ACE program emphasizes important problems in the forefront of physics and materials science, with the principle goal of investigating the fundamental microscopic mechanisms of self-organization in model complex fluid systems composed of specially synthesized colloidal particles with well-understood and controlled interactions. Our presentation affirmed the scientific and technical merit, as well as the significance of our research objectives and the need for a long-duration microgravity environment. The review panel confirmed the adequacy of the test matrix and mission success criteria. • Samples for BCAT-C1 The following samples were shipped to ZIN Technologies on October 26, 2011. NYU samples 8 – 10 are comprised of batch ADH044 PMMA particles. Samples 9 and 10 include a small concentration of spherical seed particles denoted by their respective batch numbers ADH049 and ADH050. We also prepared and characterized PMMA particles for Prof. B. Frisken & Dr. A. Bailey, Simon Fraser Univ. Approximately 21 g of PMMA particles were prepared at NYU and shipped April 25, 2012. BCAT-C1 samples 8, 9, 10, seeded growth, will build-on our previous BCAT experiments that confirmed that glasses in concentrated systems, as seen on Earth, tend to order into crystals when gravitational jamming is removed. The monodisperse particles created for this sample are the size of the wavelength of light. In this experiment, the primary particle size and concentration is fixed, while the seed size is varied. This size ratio is a sensitive parameter that affects the rate of crystal nucleation. While a significant effect is predicted theoretically, this experiment has been complicated because of particle sedimentation on earth. Thus, microgravity allows us to have both index matching (for clarity) and density matching (to avoid sedimentation in 1 g).
Bibliography Type:	Description: (Last Updated: 09/29/2025)
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